

Interior Lighting



Special Tool(s)	
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

The Body Control Module (BCM) controls all the interior lighting functions and timing by monitoring inputs from the following:

- door ajar switches
- ignition state
- instrument panel dimmer switch (monitored by the Steering Column Control Module (SCCM))
- Remote Keyless Entry (RKE) system
- vehicle speed

The BCM turns off the courtesy lamps when the vehicle speed exceeds 15 km/h (9 mph) or 5 km/h (3 mph) for the puddle lamps, unless the courtesy lamps are turned on using the instrument panel dimmer switch.

Interior Lamp Arbitrator

The interior lamp arbitrator (programming within the BCM) chooses between the interior lighting delay, the illuminated entry and exit, the battery saver, and the alarm flash processing to determine which feature has precedence of activating and deactivating the interior lamps.

Ambient Lighting

The HVAC provides voltage and ground to the LEDs. There are 3 different color LEDs (red, blue and green) housed within each LED assembly. By illuminating various color combinations, the LEDs can produce 7 different colors of ambient light.

The HVAC module receives input from the APIM and the Body Control Module (BCM) over the communication network for color and brightness settings.

The display interface in the center of the instrument panel is used to cycle through different color variations or turn the ambient lighting feature on or off. ~~APIM~~ communicates messages over the ~~HS-CAN~~ to the Body Control Module (BCM) .

The Body Control Module (BCM) gateways the messages to the HVAC module over the ~~MS-CAN~~ . The HVAC module retains the last color and brightness setting between uses.

Field-Effect Transistor (FET) Protection

A Field-Effect Transistor (FET) is a type of transistor that, when used with module software, monitors and controls current flow on module outputs. The ~~FET~~ protection strategy prevents module damage in the event of excessive current flow. For additional information, refer to Principles of Operation in Section 419-10.

The ~~BCM FET~~ protected output circuits for the interior lighting system are the interior courtesy lamp and puddle lamp output circuits.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Door ajar switches	• Body Control Module (BCM) fuse(s): ▪ 12 (15A) (courtesy lamps) ▪ 21 (10A) (ambient lighting) • Wiring, terminals, or connectors • Cargo lamp (part of the high mounted stoplamp) • Instrument panel dimmer switch • Interior lamps • Puddle lamps • Vanity mirror lamps • BCM

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove-out confirms power and ground from the ~~DLC~~ are provided to the ~~VCM~~ .*

If the scan tool does not communicate with the ~~VCM~~ :

- check the ~~VCM~~ connection to the vehicle.
- check the scan tool connection to the ~~VCM~~ .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:
 - verify the ignition is ON.

- The air bag warning indicator prove-out confirms ignition ON (other indicators may not prove ignition ON). If the ignition does not turn on, refer to Section 211-05 to diagnose no power in RUN.
 - verify the scan tool operation with a known good vehicle.
 - refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.
7. Carry out the network test.
- If the scan tool responds with no communication with one or more modules, refer to Section 418-00.
 - If the network test passes, retrieve and record the continuous memory DTCs.
8. Clear the continuous DTCs and carry out the self-test diagnostics for the BCM, the Interior Lighting Control Module (ILCM) and the Steering Column Control Module (SCCM) .
9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.
10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

Body Control Module (BCM) DTC Chart

DTC	Description	Action
B1175:01	Drivers Door Ajar Switch: General Electrical Failure	GO to Pinpoint Test F.
B1176:01	Passenger Door Ajar Switch: General Electrical Failure	GO to Pinpoint Test F.
B11C0:01	Driver Side Rear Door Ajar Switch: General Electrical Failure	GO to Pinpoint Test F.
B11C1:01	Passenger Side Rear Door Ajar Switch: General Electrical Failure	GO to Pinpoint Test F.
B12C2:11	Puddle Lamp: Circuit Short To Ground	GO to Pinpoint Test E.
B1313:11	Interior Lighting Output: Circuit Short To Ground	GO to Pinpoint Test C.
B1313:15	Interior Lighting Output: Circuit Short To Battery or Open	If the interior courtesy lamps are inoperative, GO to Pinpoint Test C. If the interior courtesy lamps stay on continuously, GO to Pinpoint Test G.
U1000:00	Solid State Driver Protection Active - Driver Disabled: No Sub Type Information	The module has temporarily disabled an output because an excessive current draw exists (such as a short to ground). The Body Control Module (BCM) cannot enable the output until the cause of the short is corrected. ADDRESS all other DTCs first. After the cause of the concern is corrected, CLEAR the DTCs. REPEAT the self-test.
U3000:49	Control Module: Internal Electronic Failure	The module has permanently disabled an output because an excessive current draw fault (such as a short to ground) has exceeded the limits that the BCM can withstand. The cause of the excessive current draw MUST be corrected before a new BCM is installed. ADDRESS all other DTCs first. After the cause of the

DTC	Description	Action
		concern is corrected, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
All other DTCs	—	REFER to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

Steering Column Control Module (SCCM) DTC Chart

DTC	Description	Action
B1124:09	Lamp Fade Control: Component Failure	REFER to Section 413-00.
All other DTCs	—	REFER to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

HVAC Module DTC Chart

DTC	Description	Action
B127C:11	Ambient Lighting Zone 2 Output Red LED: Circuit Short To Ground	GO to Pinpoint Test L.
B127C:12	Ambient Lighting Zone 2 Output Red LED: Circuit Short To Battery	GO to Pinpoint Test L.
B127C:13	Ambient Lighting Zone 2 Output Red LED: Circuit Open	GO to Pinpoint Test L.
B127F:11	Ambient Lighting Zone 2 Output Green LED: Circuit Short To Ground	GO to Pinpoint Test L.
B127F:12	Ambient Lighting Zone 2 Output Green LED: Circuit Short To Battery	GO to Pinpoint Test L.
B127F:13	Ambient Lighting Zone 2 Output Green LED: Circuit Open	GO to Pinpoint Test L.
B1282:11	Ambient Lighting Zone 2 Output Blue LED: Circuit Short To Ground	GO to Pinpoint Test L.
B1282:12	Ambient Lighting Zone 2 Output Blue LED: Circuit Short To Battery	GO to Pinpoint Test L.
B1282:13	Ambient Lighting Zone 2 Output Blue LED: Circuit Open	GO to Pinpoint Test L.
All other DTCs	—	REFER to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

Symptom Chart

Symptom Chart

Condition	Possible Causes	Action
The courtesy lamps flash on and off	The Body Control Module (BCM) is not configured correctly	CARRY OUT the Programmable Module Installation (PMI) for the BCM . REFER to Section 418-01. TEST the system for normal operation.
The courtesy lamps do not illuminate with a door open	- Wiring, terminals or connectors - Door ajar switch - BCM	GO to Pinpoint Test A.
The courtesy lamps do not illuminate from the instrument panel dimmer switch	- Wiring, terminals or connectors - Instrument panel dimmer switch - Steering Column Control Module (SCCM)	GO to Pinpoint Test B.
One or more interior courtesy lamp is inoperative	- Fuse - Wiring, terminals or connectors - Interior lamp - BCM	VERIFY the bulb(s) is OK. If OK, GO to Pinpoint Test C.
One or more puddle lamp is inoperative	- Wiring, terminals or connectors - Puddle lamp - Exterior mirror	VERIFY the bulb(s) is OK. If OK, GO to Pinpoint Test D.
The cargo lamp is inoperative	- Fuse - Wiring, terminals or connectors - High mounted stoplamp - BCM	VERIFY the bulb(s) is OK. If OK, GO to Pinpoint Test E.
All the courtesy lamps stay on continuously	- Wiring, terminals or connectors - Door ajar switch	GO to Pinpoint Test F.

	• BCM	
• The interior courtesy lamps stay on continuously	• Wiring, terminals or connectors • BCM	• GO to Pinpoint Test G.
• The cargo lamp/puddle lamps stay on continuously	• Wiring, terminals or connectors • BCM	• GO to Pinpoint Test H.
• One or more demand lamp is inoperative	• Fuse • Wiring, terminals or connectors • Sun visor • Interior lamp • BCM	• VERIFY the bulb(s) is OK. If OK, GO to Pinpoint Test I.
• The battery saver does not deactivate after time-out	• Wiring, terminals or connectors • BCM	• GO to Pinpoint Test J.
• An individual ambient lighting LED is inoperative/does not cycle through all color combinations	• Ambient lighting harness	• NOTE: <i>The LEDs are not serviceable separate from the harness.</i> • REPAIR or INSTALL a new ambient lighting harness in question. TEST the system for normal operation.
• The ambient lighting is inoperative	• Wiring, terminals or connectors • HVAC module	• GO to Pinpoint Test K.
• The ambient lighting does not cycle through all color combinations	• Wiring, terminals or connectors • HVAC module	• GO to Pinpoint Test L.

Pinpoint Tests

Pinpoint Test A: The Courtesy Lamps Do Not Illuminate With A Door Open

Refer to Wiring Diagrams Cell [117](#) , Remote Keyless Entry and Alarm for schematic and connector information.

Normal Operation

The Body Control Module (BCM) sends a voltage signal to each door ajar switch, which is routed to ground when the door is closed (the door ajar switch is closed when the door is closed). When the door is opened, the signal to ground is

interrupted indicating an open door to the BCM .

When the BCM detects an open door, voltage is provided to the courtesy lamps.

NOTE: All the door ajar switches are integral to the door latch **except** for the rear door ajar switches on Super Cab vehicles. The rear door ajar switches on Super Cab vehicles are replaced independently of the door latch.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Door ajar switch
- BCM

PINPOINT TEST A : THE COURTESY LAMPS DO NOT ILLUMINATE WITH A DOOR OPEN

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 CHECK THE OPERATION OF THE COURTESY LAMPS FROM THE INSTRUMENT PANEL DIMMER SWITCH

- Ignition OFF.
- Press and hold the instrument panel dimmer switch to the DOME LAMP ON (second detent up) position for 3 seconds.

Do the interior courtesy lamps illuminate?

Yes	GO to A2 .
No	GO to Pinpoint Test C .

A2 CHECK THE DOOR AJAR SWITCH OPERATION

- Press and hold the instrument panel dimmer switch to the DOME LAMP OFF (second detent down) position for 3 seconds.
- Disconnect: Suspect Door Ajar Switch .

Do the courtesy lamps illuminate?

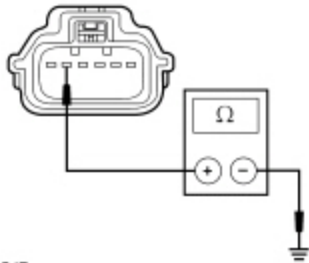
Yes	INSTALL a new door latch or ajar switch in question. REFER to Section 501-14. TEST the system for normal operation.
No	GO to A3 .

A3 CHECK THE DOOR AJAR SWITCH INPUT CIRCUIT FOR A SHORT TO GROUND

- Disconnect: BCM [C2280C](#) .
- For any door except a rear door on Super Cab vehicles, measure the resistance between the suspect door latch, harness side and ground as follows:

Location	Connector-Pin	Circuit
LH front	C525 Pin 5	CPL26 (GN/VT)

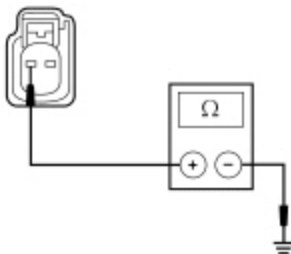
Location	Connector-Pin	Circuit
RH front	C609 Pin 5	CPL31 (WH)
LH rear	C705 Pin 5	CPL36 (GN)
RH rear	C805 Pin 5	CPL39 (YE)



• N0090547

- For a rear door on Super Cab vehicles, measure the resistance between the suspect door ajar switch, harness side and ground as follows:

Location	Connector-Pin	Circuit
LH rear	C715 Pin 2	CPL36 (GN)
RH rear	C820 Pin 2	CPL39 (YE)



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Is the resistance greater than 10,000 ohms?

Yes	GO to A4 .
No	REPAIR the circuit in question for a short to ground. TEST the system for normal operation.

A4 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.

- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test B: The Courtesy Lamps Do Not Illuminate From The Instrument Panel Dimmer Switch

Refer to Wiring Diagrams Cell [89](#) , Interior Lighting for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) sends 3 voltage signals to the instrument panel dimmer switch to monitor the DIM UP, DIM DOWN and DOME LAMP request inputs. The SCCM sends a message over the communication network to the Body Control Module (BCM) to indicate the instrument panel dimmer switch position.

The BCM turns the courtesy lamps on and off by monitoring the dimmer switch and dome lamp request input circuits. When the instrument panel dimmer switch is pressed to the second detent up, the SCCM receives signals indicating DIM UP and DOME LAMP request circuits at the same time, indicating a request to turn the courtesy lamps on. When the instrument panel dimmer switch is pressed to the second detent down, the SCCM receives signals indicating DIM DOWN and DOME LAMP request circuits at the same time, indicating a request to turn the courtesy lamps off. When the SCCM detects these inputs from the instrument panel dimmer switch, the SCCM sends the appropriate message to the BCM and voltage is supplied to the courtesy lamps.

If the SCCM detects a fault with any of the 3 inputs from the switch (circuit shorted to ground for example), the SCCM sets DTC B1124:09 (Lamp Fade Control: Component Failure). The BCM then defaults the dimmable backlighting to a preset level that cannot be adjusted.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Instrument panel dimmer switch
- SCCM

PINPOINT TEST B : THE COURTESY LAMPS DO NOT ILLUMINATE FROM THE INSTRUMENT PANEL DIMMER SWITCH

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
B1 CHECK THE OPERATION OF THE COURTESY LAMPS FROM A DOOR	
• Place the headlamp switch in the OFF position. • Open any door and observe the interior courtesy lamps.	
Do the interior courtesy lamps illuminate?	
Yes	GO to B2 .
No	GO to Pinpoint Test C .

B2 CHECK THE DIMMABLE BACKLIGHTING OPERATION

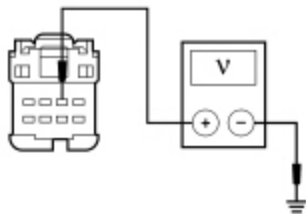
- Place the headlamp switch in the PARKING LAMPS ON position.
- Monitor the instrument panel dimmable backlighting while pressing (and holding) the instrument panel dimmer switch to the first detent up and then the first detent down.

Does the dimmable backlighting intensity change with the instrument panel dimmer switch position?

Yes	GO to B3 .
No	REFER to Section 413-00.

B3 CHECK FOR A VOLTAGE SIGNAL TO THE INSTRUMENT PANEL DIMMER SWITCH

- Ignition OFF.
- Disconnect: Instrument Panel Dimmer Switch [C2298](#).
- Measure the voltage between the instrument panel dimmer switch [C2298](#) Pin 2, circuit VLN36 (GY/OG), harness side and ground.



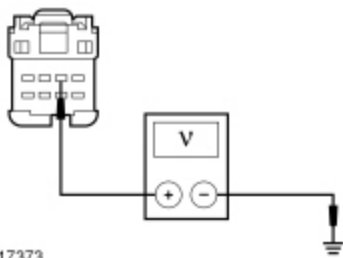
• N0102465

Is the voltage approximately 5 volts?

Yes	INSTALL a new instrument panel dimmer switch. REFER to Section 413-00. TEST the system for normal operation.
No	GO to B4 .

B4 CHECK THE DOME LAMP REQUEST INPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: SCCM [C2414A](#) .
- Ignition ON.
- Measure the voltage between the instrument panel dimmer switch [C2298](#) Pin 2, circuit VLN36 (GY/OG), harness side and ground.



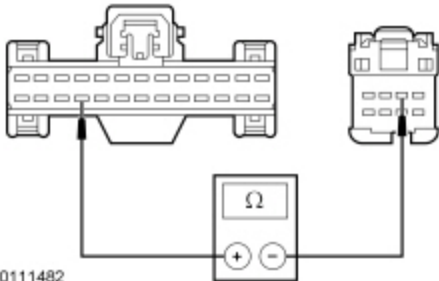
N0117373

Is any voltage present?

Yes	REPAIR circuit VLN36 (GY/OG) for a short to voltage. TEST the system for normal operation.
No	GO to B5 .

B5 CHECK THE DOME LAMP REQUEST INPUT CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the SCCM [C2414A](#) Pin 23, circuit VLN36 (GY/OG), harness side and the instrument panel dimmer switch [C2298](#) Pin 2, circuit VLN36 (GY/OG), harness side.



N0111482

Is the resistance less than 5 ohms?

Yes	GO to B6 .
No	REPAIR circuit VLN36 (GY/OG) for an open. TEST the system for normal operation.

B6 CHECK FOR CORRECT SCCM OPERATION

- Disconnect all the SCCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the SCCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Section 211-05. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test C: One Or More Interior Courtesy Lamp Is Inoperative

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Normal Operation

The Body Control Module (BCM) fuse 12 (15A) supplies power for the interior courtesy lamps (and the running board lamps if equipped). When the BCM receives the required input (such as an open door or request from the instrument panel dimmer switch), it provides voltage to the interior courtesy lamps.

DTC Description	Fault Trigger Conditions
B1313:11 — Interior Lighting Output: Circuit Short To Ground	This DTC sets when the BCM detects a short to ground from the courtesy lighting output circuit.
B1313:15 — Interior Lighting Output: Circuit Short To Battery or Open	This DTC sets when the BCM detects an open from the courtesy lighting output circuit.
U1000:00 — Solid State Driver Protection Active - Driver Disabled: No Sub Type Information	This DTC sets when the BCM has temporarily shut down the output driver. The module has temporarily disabled the interior courtesy lamp output because an excessive current draw exists (such as a short to ground). The BCM cannot enable the courtesy lamp output until the cause of the short is corrected, the DTCs have been cleared and a successful self-test is run.
U3000:49 — Control Module: Internal Electronic Failure	This DTC sets when the BCM has permanently shut down the output driver. The module has permanently disabled the interior courtesy lamp output because an excessive current draw fault (such as a short to ground) has exceeded the limits that the BCM can withstand. The cause of the excessive current draw MUST be corrected before a new BCM is installed.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Interior lamp
- BCM

PINPOINT TEST C : ONE OR MORE INTERIOR COURTESY LAMP IS INOPERATIVE
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
C1 CHECK THE OPERATION OF ALL THE INTERIOR COURTESY LAMPS

- Press and hold the instrument panel dimmer switch to the DOME LAMP ON (second detent up) position for 3 seconds.

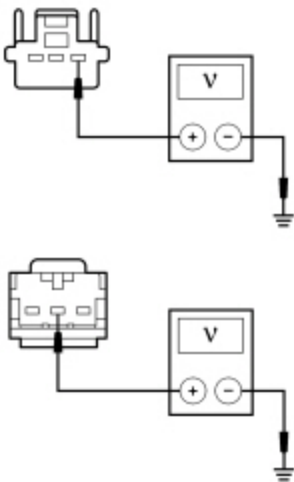
Are all the interior courtesy lamps inoperative?

Yes	VERIFY the BCM fuse 12 (15A) is OK. If OK, GO to C4. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. After the repair: If no DTCs are present, TEST the system for normal operation. If DTC U1000:00 is present, CLEAR the DTCs and REPEAT the self-test (required to enable the courtesy lamp output driver if DTC U1000:00 is present). TEST the system for normal operation. If DTC U3000:49 is present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	GO to C2 .

C2 CHECK FOR VOLTAGE TO THE INTERIOR LAMP

- Turn the interior courtesy lamps off.
- Disconnect: Inoperative Interior Lamp .
- Press and hold the instrument panel dimmer switch to the DOME LAMP ON (second detent up) position for 3 seconds.
- Measure the voltage between the inoperative interior lamp, harness side and ground as follows:

Inoperative Lamp	Connector-Pin	Circuit
Front interior lamp (without overhead console)	C901 Pin 1	VLN33 (GY/VT)
Front interior lamp (with overhead console)	C929 Pin 2	VLN33 (GY/VT)
Rear interior lamp	C932 Pin 1	VLN33 (GY/VT)



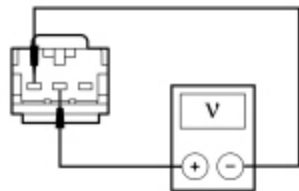
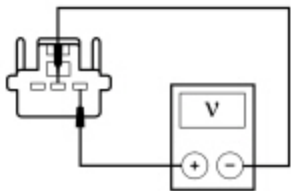
Is the voltage greater than 10 volts?

Yes	GO to C3 .
No	REPAIR circuit VLN33 (GY/VT) for an open. TEST the system for normal operation.

C3 CHECK THE INTERIOR LAMP GROUND CIRCUIT FOR AN OPEN

- Measure the voltage between the suspect interior lamp pins, harness side as follows:

Inoperative Lamp	Connector-Pin Circuit	Connector-Pin Circuit
Front interior lamp (without overhead console)	C901 Pin 1 VLN33 (GY/VT)	C901 Pin 2 GD139 (BK/YE)
Front interior lamp (with overhead console)	C929 Pin 2 VLN33 (GY/VT)	C929 Pin 1 GD139 (BK/YE)
Rear interior lamp	C932 Pin 1 VLN33 (GY/VT)	C932 Pin 2 GD139 (BK/YE)



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Is the voltage greater than 10 volts?

Yes	INSTALL a new interior lamp for the lamp in question. REFER to Interior Lamp in this section. TEST the system for normal operation.
No	REPAIR circuit GD139 (BK/YE) for an open. TEST the system for normal operation.

C4 CHECK THE OPERATION OF THE DEMAND LIGHTING

- Operate the demand lighting function (such as the map light) on any interior lamp.

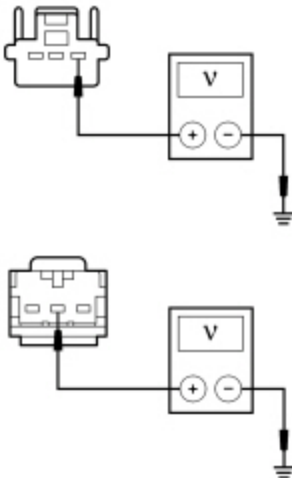
Does the interior lamp illuminate?

Yes	For vehicles that only have a single interior lamp, GO to C5 . For vehicles that have 2 interior lamps, GO to C6 .
No	REPAIR circuit GD139 (BK/YE) for an open. TEST the system for normal operation.

C5 CHECK FOR VOLTAGE TO THE INTERIOR LAMP

- Press and hold the instrument panel dimmer switch to the DOME LAMP OFF (second detent down) position for 3 seconds.
- Disconnect: Front Interior Lamp .
- Ignition ON.
- Press and hold the instrument panel dimmer switch to the DOME LAMP ON (second detent up) position for 3 seconds.
- Measure the voltage between the interior lamp, harness side and ground as follows:

Inoperative Lamp	Connector-Pin	Circuit
Front interior/map lamp (without overhead console)	C901 Pin 1	VLN33 (GY/VT)
Front interior/map lamp (with overhead console)	C929 Pin 2	VLN33 (GY/VT)



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Is the voltage greater than 10 volts?

Yes	INSTALL a new interior lamp. REFER to Interior Lamp in this section. TEST the system for normal operation.
No	GO to C6 .

C6 BYPASS THE BCM

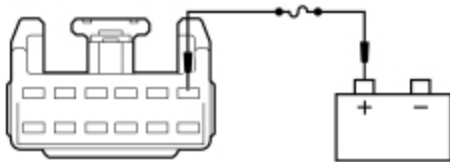
NOTE: If the jumper wire fails, refer to the Wiring Diagrams manual to identify the possible causes of the circuit short. After the repair:

If no DTCs are present, test the system for normal operation.

If DTC U1000:00 is present, clear the DTCs and repeat the self-test (**required to enable the courtesy lamp output driver if DTC U1000:00 is present**). Test the system for normal operation.

If DTC U3000:49 is present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . Test the system for normal operation.

- Connect: Front Interior Lamp .
- Disconnect: BCM [C2280A](#) .
- Connect a fused jumper wire between the BCM [C2280A](#) Pin 1, circuit VLN33 (GY/VT), harness side and battery positive.



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Do the courtesy lamps illuminate?

Yes	REMOVE the jumper wire. GO to C7 .
No	REMOVE the jumper wire. REPAIR circuit VLN33 (GY/VT) for an open. TEST the system for normal operation.

C7 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test D: One Or More Puddle Lamp Is Inoperative

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Normal Operation

The puddle lamps illuminate when the cargo lamp illuminates. The Body Control Module (BCM) uses a separate output circuit independent of the interior courtesy lamps that supplies voltage to the cargo lamp and puddle lamps.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Puddle lamp
- Exterior mirror

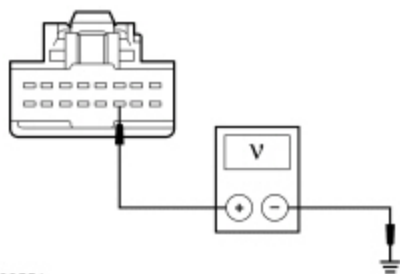
PINPOINT TEST D : ONE OR MORE PUDDLE LAMP IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
NOTE: <i>Make sure the interior courtesy lamps operate before continuing diagnostics.</i>	
D1 DETERMINE IF THE CARGO LAMP IS INOPERATIVE	
• Ignition OFF. • Press and hold the instrument panel dimmer switch to the DOME LAMP ON (second detent up) position for 3 seconds.	
Is the cargo lamp illuminated?	
Yes	GO to D2 .
No	GO to Pinpoint Test E .

D2 CHECK FOR VOLTAGE TO THE EXTERIOR MIRROR

NOTE: <i>Make sure the suspect puddle lamp bulb is OK before continuing diagnostics.</i>	
• Press and hold the instrument panel dimmer switch to the DOME LAMP OFF (second detent down) position for 3 seconds. • Disconnect: Suspect Exterior Mirror . • Press and hold the instrument panel dimmer switch to the DOME LAMP ON (second detent up) position for 3 seconds. • Measure the voltage between the LH exterior mirror connector C521 Pin 11, circuit CLN25 (VT), harness side and ground; or between the RH exterior mirror connector C615 Pin 11, circuit CLN25 (VT), harness side and	

ground.



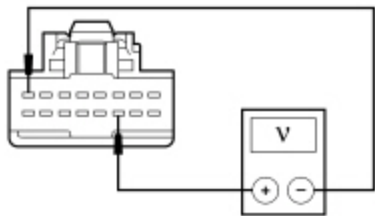
N0090551

Is the voltage greater than 10 volts?

Yes	GO to D3 .
No	REPAIR circuit CLN25 (VT) for an open. TEST the system for normal operation.

D3 CHECK THE PUDDLE LAMP GROUND CIRCUIT FOR AN OPEN

- Measure the voltage between the LH exterior mirror connector [C521](#) Pin 11, circuit CLN25 (VT), harness side and the LH exterior mirror connector [C521](#) Pin 8, circuit GD134 (BK/VT), harness side; or between the RH exterior mirror connector [C615](#) Pin 11, circuit CLN25 (VT), harness side and the RH exterior mirror connector [C615](#) Pin 8, circuit GD139 (BK/YE), harness side.



N0090552

Is the voltage greater than 10 volts?

Yes	INSTALL a new exterior mirror in question. REFER to Section 501-09. TEST the system for normal operation.
No	For the LH exterior mirror, REPAIR circuit GD134 (BK/VT) for an open. TEST the system for normal operation. For the RH exterior mirror, REPAIR circuit GD139 (BK/YE) for an open. TEST the system for normal operation.

Pinpoint Test E: The Cargo Lamp is Inoperative

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Normal Operation

The cargo lamp illuminates when the courtesy lamps illuminate. The Body Control Module (BCM) uses a separate output circuit that supplies voltage to the cargo lamp. The cargo lamp shares the ground circuit with the high mounted stoplamp.

The puddle lamps (if equipped) share the voltage supply with the cargo lamp.

DTC Description	Fault Trigger Conditions
B12C2:11 — Puddle Lamp: Circuit Short To Ground	This DTC sets when the BCM detects a short to ground from the cargo lamp output circuit.
U1000:00 — Solid State Driver Protection Active - Driver Disabled: No Sub Type Information	A continuous DTC that sets when the BCM has temporarily shut down the output driver. The module has temporarily disabled the cargo lamp output because an excessive current draw exists (such as a short to ground). The BCM cannot enable the cargo lamp output until the cause of the short is corrected, the DTCs have been cleared and a successful self-test is run.
U3000:49 — Control Module: Internal Electronic Failure	A continuous DTC that sets when the BCM has permanently shut down the output driver. The module has permanently disabled the cargo lamp output because an excessive current draw fault (such as a short to ground) has exceeded the limits that the BCM can withstand. The cause of the excessive current draw MUST be corrected before a new BCM is installed.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- High mounted stoplamp
- BCM

PINPOINT TEST E : THE CARGO LAMP IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

E1 CHECK THE OPERATION OF THE HIGH MOUNTED STOPLAMP

- Apply the brake pedal.

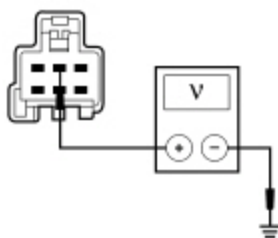
Does the high mounted stoplamp illuminate?

Yes	GO to E2 .
No	REFER to Section 417-01.

E2 CHECK FOR VOLTAGE TO THE CARGO LAMP

- Ignition OFF.
- Disconnect: High Mounted Stoplamp [C904](#).
- Press and hold the instrument panel dimmer switch to the DOME LAMP ON (second detent up) position for 3 seconds.

- Measure the voltage between the high mounted stoplamp [C904](#) Pin 2, circuit CLN25 (VT), harness side and ground.



• N0104860

Is the voltage greater than 10 volts?

Yes	INSTALL a new high mounted stoplamp. REFER to Section 417-01. TEST the system for normal operation.
No	GO to E3 .

E3 BYPASS THE BCM

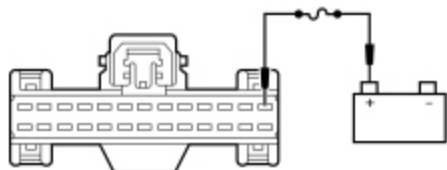
NOTE: If the jumper wire fails, refer to the Wiring Diagrams manual to identify the possible causes of the circuit short. After the repair:

If no DTCs are present, test the system for normal operation.

If DTC U1000:00 is present, clear the DTCs and repeat the self-test (**required to enable the cargo lamp output driver if DTC U1000:00 is present**). Test the system for normal operation.

If DTC U3000:49 is present, **INSTALL** a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . Test the system for normal operation.

- Press and hold the instrument panel dimmer switch to the DOME LAMP OFF (second detent down) position for 3 seconds.
- Connect: High Mounted Stoplamp [C904](#).
- Disconnect: BCM [C2280C](#) .
- Connect a fused jumper wire between the BCM [C2280C](#) Pin 1, circuit CLN25 (VT), harness side and battery positive.



• N0111485

Does the cargo lamp illuminate?

Yes	REMOVE the jumper wire. GO to E4 .
------------	--

No	REMOVE the jumper wire. REPAIR circuit CLN25 (VT) for an open. TEST the system for normal operation.
-----------	--

E4 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test F: All The Courtesy Lamps Stay On Continuously

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Refer to Wiring Diagrams Cell [117](#) , Remote Keyless Entry and Alarm for schematic and connector information.

Normal Operation

The Body Control Module (BCM) sends a voltage signal to each door ajar switch, which is routed to ground when the door is closed (the door ajar switch is closed when the door is closed). When the door is opened, the signal to ground is interrupted indicating an open door to the **BCM** .

NOTE: All the door ajar switches are integral to the door latch ***except*** for the rear door ajar switches on Super Cab vehicles. The rear door ajar switches on Super Cab vehicles are replaced independently of the door latch.

When the **BCM** detects an open door or a request from the instrument panel dimmer switch, voltage is provided to the courtesy lamps.

DTC Description	Fault Trigger Conditions
• B1175:01 — Drivers Door Ajar Switch: General Electrical Failure	This DTC sets when the BCM detects an open from the LH front door ajar circuit.

DTC Description	Fault Trigger Conditions
B1176:01 — Passenger Door Ajar Switch: General Electrical Failure	This DTC sets when the BCM detects an open from the RH front door ajar circuit.
B11C0:01 — Driver Side Rear Door Ajar Switch: General Electrical Failure	This DTC sets sets when the BCM detects an open from the LH rear door ajar circuit.
B11C1:01 — Passenger Side Rear Door Ajar Switch: General Electrical Failure	This DTC sets when the BCM detects an open from the RH rear door ajar circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Door ajar switch
- BCM

PINPOINT TEST F : ALL THE COURTESY LAMPS STAY ON CONTINUOUSLY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

F1 CHECK THE DOOR AJAR SWITCH PIDS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: BCM DataLogger .
- Monitor the BCM door ajar switch PIDs (DOOR_SW_DRVR, DOOR_SW_PSGR, DOOR_SW_LR, DOOR_SW_RR) while opening and closing all the doors.

Do all the door ajar switch PID values agree with the door positions?

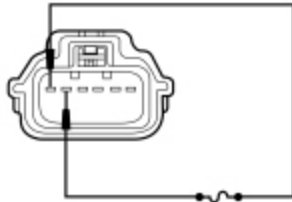
Yes	If the interior courtesy lamps are on, GO to Pinpoint Test G . If the cargo lamp or puddle lamps are on, GO to Pinpoint Test H .
No	GO to F2 .

F2 BYPASS THE SUSPECT DOOR AJAR SWITCH

- Disconnect: Suspect Door Ajar Switch .
- For any door except a rear door on Super Cab vehicles, connect a fused jumper wire between the suspect door ajar switch connector pins, harness side as follows:

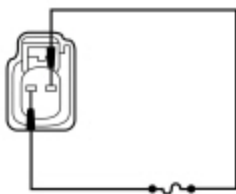
Suspect Door Ajar Switch	Connector-Pin Circuit	Connector-Pin Circuit
LH front	C525 Pin 5 CPL26 (GN/VT)	C525 Pin 6 GD134 (BK/VT)
RH front	C609 Pin 5	C609 Pin 6

Suspect Door Ajar Switch	Connector-Pin Circuit	Connector-Pin Circuit
	CPL31 (WH)	GD139 (BK/YE)
LH rear	C705 Pin 5 CPL36 (GN)	C705 Pin 6 GD134 (BK/VT)
RH rear	C805 Pin 5 CPL39 (YE)	C805 Pin 6 GD139 (BK/YE)



- N0119594
- For a rear door on Super Cab vehicles, connect a fused jumper wire between the suspect door ajar switch connector pins, harness side as follows:

Location	Connector-Pin Circuit	Connector-Pin Circuit
LH rear	C715 Pin 2 CPL36 (GN)	C715 Pin 1 GD134 (BK/VT)
RH rear	C820 Pin 2 CPL39 (YE)	C820 Pin 1 GD139 (BK/YE)



- N0119593
- Enter the following diagnostic mode on the scan tool: **BCM** DataLogger .
- Monitor the suspect **BCM** door ajar switch PID (DOOR_SW_DRVR, DOOR_SW_PSGR, DOOR_SW_LR, DOOR_SW_RR).

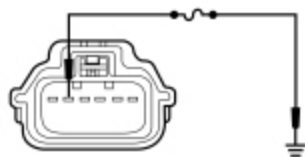
Does the PID indicate the door is closed?

Yes	REMOVE the jumper wire. INSTALL a new door latch or door ajar switch in question. REFER to Section 501-14. TEST the system for normal operation.
No	REMOVE the jumper wire. GO to F3 .

F3 BYPASS THE DOOR AJAR SWITCH GROUND CIRCUIT

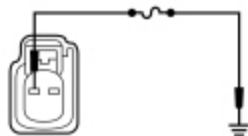
- For any door except a rear door on Super Cab vehicles, connect a fused jumper wire between the suspect door ajar switch connector, harness side and ground as follows:

Suspect Door Ajar Switch	Connector-Pin	Circuit
LH front	C525 Pin 5	CPL26 (GN/VT)
RH front	C609 Pin 5	CPL31 (WH)
LH rear	C705 Pin 5	CPL36 (GN)
RH rear	C805 Pin 5	CPL39 (YE)



- N0119595
- For a rear door on Super Cab vehicles, connect a fused jumper wire between the suspect door ajar switch connector, harness side and ground as follows:

Location	Connector-Pin	Circuit
LH rear	C715 Pin 2	CPL36 (GN)
RH rear	C820 Pin 2	CPL39 (YE)



- N0119596
- Enter the following diagnostic mode on the scan tool: **BCM DataLogger** .
- Monitor the suspect **BCM** door ajar switch PID (DOOR_SW_DRV, DOOR_SW_PSGR, DOOR_SW_LR, DOOR_SW_RR).

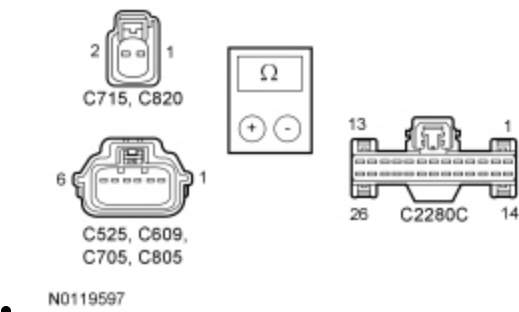
Does the PID indicate the door is closed?

Yes	REMOVE the jumper wire. REPAIR the ground circuit in question for an open. TEST the system for normal operation.
No	REMOVE the jumper wire. GO to F4 .

F4 CHECK THE SUSPECT DOOR AJAR SWITCH INPUT CIRCUIT FOR AN OPEN

- Disconnect: BCM [C2280C](#) .
- Measure the resistance between the suspect door ajar switch connector, harness side and the BCM connector, harness side as follows:

Suspect Door Ajar Switch Connector-Pin	BCM Connector-Pin	Circuit
LH front C525 Pin 5	C2280C Pin 9	CPL26 (GN/VT)
RH front C609 Pin 5	C2280C Pin 5	CPL31 (WH)
LH rear (Super Crew) C705 Pin 5	C2280C Pin 20	CPL36 (GN)
LH rear (Super Cab) C715 Pin 2	C2280C Pin 20	CPL36 (GN)
RH rear (Super Crew) C805 Pin 5	C2280C Pin 21	CPL39 (YE)
RH rear (Super Crew) C820 Pin 2	C2280C Pin 21	CPL39 (YE)



Is the resistance less than 5 ohms?

Yes	GO to F5 .
No	REPAIR the ajar input circuit in question for an open. TEST the system for normal operation.

F5 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.

- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test G: The Interior Courtesy Lamps Stay On Continuously

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Normal Operation

When the Body Control Module (BCM) detects a request for the courtesy lamps (such as an open door or from the instrument panel dimmer switch), the BCM provides voltage to the interior courtesy lamps.

- DTC B1313:15 (Interior Lighting Output: Circuit Short To Battery or Open) — this DTC sets when the BCM detects a short to voltage from the interior courtesy lighting output circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- BCM

PINPOINT TEST G : THE INTERIOR COURTESY LAMPS STAY ON CONTINUOUSLY	
NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
G1 CHECK THE DOOR AJAR SWITCH PIDS	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: BCM DataLogger . • Monitor the BCM door ajar switch PIDs (DOOR_SW_DRVR, DOOR_SW_PSGR, DOOR_SW_LR, DOOR_SW_RR) while opening and closing all the doors.	
Do all the door ajar switch PID values agree with the door positions?	
Yes	GO to G2 .
No	GO to Pinpoint Test F .
G2 CHECK THE INTERIOR COURTESY LAMP OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE	
• Ignition OFF. • Disconnect: BCM C2280A . • Ignition ON.	

Do the interior courtesy lamps continue to illuminate?

Yes	REPAIR circuit VLN33 (GY/VT) for a short to voltage. TEST the system for normal operation.
No	GO to G3 .

G3 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test H: The Cargo Lamp/Puddle Lamps Stay On Continuously

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Normal Operation

When the Body Control Module (BCM) detects a request for the courtesy lamps (such as an open door or from the instrument panel dimmer switch), the BCM provides voltage to the cargo lamp and puddle lamps (if equipped).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- BCM

PINPOINT TEST H : THE CARGO LAMP/PUDDLE LAMPS STAY ON CONTINUOUSLY

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

H1 CHECK THE INTERIOR COURTESY LAMPS

- Observe the interior courtesy lamps.

Are the interior courtesy lamps illuminated?

Yes	GO to Pinpoint Test F .
No	GO to H2 .

H2 CHECK THE CARGO LAMP OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM [C2280C](#) .
- Ignition ON.

Does the cargo lamp continue to illuminate?

Yes	REPAIR circuit CLN25 (VT) for a short to voltage. TEST the system for normal operation.
No	GO to H3 .

H3 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test I: One Or More Demand Lamp Is Inoperative

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Normal Operation

The Body Control Module (BCM) controls the operation of the battery saver relay. The BCM deactivates the battery saver relay 10 minutes after the ignition is switched to off and no other inputs are detected by the BCM to energize the battery saver relay (such as a flash-to-pass request or a button is pressed on a Remote Keyless Entry (RKE) transmitter).

When the battery saver relay is energized, the BCM supplies voltage to the demand lamps.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Sun visor
- Interior lamp
- BCM

PINPOINT TEST I : ONE OR MORE DEMAND LAMP IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

I1 CHECK ALL THE DEMAND LAMPS

- Ignition OFF.
- Check the operation of all the interior demand lamps (map lamps and vanity mirror lamps).

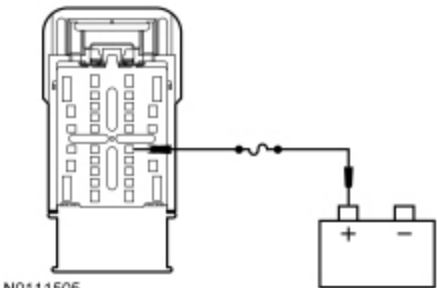
Are all the interior demand lamps inoperative?

Yes	GO to I2 .
No	GO to I3 .

I2 BYPASS THE BCM

NOTE: *If the fuse fails, repair circuit CLN09 (YE/GN) for a short to ground.*

- Disconnect: ~~BCM~~ [C2280D](#) .
- Connect a fused jumper wire between the ~~BCM~~ [C2280D](#) Pin 22, circuit CLN09 (YE/GN), harness side and battery positive.



N0111505

- Switch any demand lamp (map or vanity mirror) on.

Does the demand lamp illuminate when switched on?

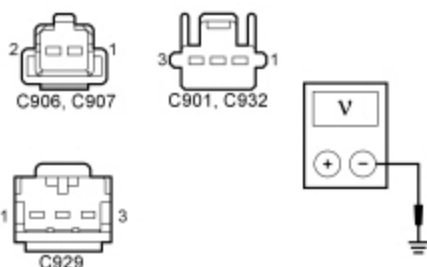
Yes	REMOVE the jumper wire. VERIFY the BCM fuse 4 (10A) is OK. If OK, GO to I5 . If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short.
No	REMOVE the jumper wire. REPAIR circuit CLN09 (YE/GN) for an open. TEST the system for normal operation.

I3 CHECK FOR VOLTAGE TO THE INOPERATIVE LAMP

- Ignition OFF.
- Disconnect: Inoperative Lamp .

- Ignition ON.
- Measure the voltage between the inoperative demand lamp, harness side and ground as follows:

Inoperative Lamp	Connector-Pin	Circuit
LH vanity mirror	C907 Pin 1	CLN09 (YE/GN)
RH vanity mirror	C906 Pin 1	CLN09 (YE/GN)
Front interior/map (without overhead console)	C901 Pin 3	CLN09 (YE/GN)
Front interior/map (with overhead console)	C929 Pin 3	CLN09 (YE/GN)
Rear interior/map	C932 Pin 3	CLN09 (YE/GN)



N0117349

Is the voltage greater than 10 volts?

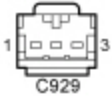
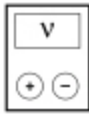
Yes	GO to I4 .
No	REPAIR circuit CLN09 (YE/GN) for an open. TEST the system for normal operation.

I4 CHECK THE INOPERATIVE LAMP GROUND CIRCUIT FOR AN OPEN

- Measure the voltage between the inoperative lamp connector pins, harness side as follows:

Inoperative Lamp	Connector-Pin Circuit	Connector-Pin Circuit
LH vanity mirror	C907 Pin 1 CLN09 (YE/GN)	C907 Pin 2 GD139 (BK/YE)
RH vanity mirror	C906 Pin 1 CLN09 (YE/GN)	C906 Pin 2 GD139 (BK/YE)
Front interior (without overhead console)	C901 Pin 3 CLN09 (YE/GN)	C901 Pin 2 GD139 (BK/YE)
Front interior (with overhead console)	C929 Pin 3 CLN09 (YE/GN)	C929 Pin 1 GD139 (BK/YE)
Rear interior	C932 Pin 3	C932 Pin 2

Inoperative Lamp	Connector-Pin	Connector-Pin
	Circuit	Circuit
	CLN09 (YE/GN)	GD139 (BK/YE)



N0117350

Is the voltage greater than 10 volts?

Yes	INSTALL a new lamp for the lamp in question. REFER to Interior Lamp in this section. TEST the system for normal operation.
No	REPAIR the ground circuit in question for an open. TEST the system for normal operation.

15 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the **BCM** connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the **BCM** connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test J: The Battery Saver Does Not Deactivate After Time-Out

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Normal Operation

The Body Control Module (BCM) controls the operation of the battery saver relay. The BCM deactivates the battery saver relay 10 minutes after the ignition is switched off and no other inputs are detected by the BCM to energize the battery saver relay (such as a flash-to-pass request or a button is pressed on a Remote Keyless Entry (RKE) transmitter).

When the battery saver relay is energized, the BCM supplies voltage to the demand lamps.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- BCM

PINPOINT TEST J : THE BATTERY SAVER DOES NOT DEACTIVATE AFTER TIME-OUT

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

J1 CHECK THE INTERIOR DEMAND LAMP VOLTAGE SUPPLY CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM [C2280D](#) .
- Ignition ON.
- Turn any demand lamp (map or vanity mirror) on.

Does the demand lamp turn on?

Yes	REPAIR circuit CLN09 (YE/GN) for a short to voltage. TEST the system for normal operation.
No	GO to J2 .

J2 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Normal Operation

The HVAC module provides voltage and ground to the LEDs. There are 3 LEDs (red, blue and green) housed within each LED assembly. By illuminating various color combinations, the LEDs are able to produce 7 different colors of ambient light.

The HVAC module receives input from the Accessory Protocol Interface Module (APIM) and the Body Control Module (BCM) over the communication network for color and brightness settings. The HVAC module retains the last color and brightness setting between uses.

There ambient lighting system illuminates the following:

- Cupholders (located in the center console)
- Center console storage compartment
- Front footwells
- Rear footwells

The HVAC module has a separate ground circuit for the ambient lighting system.

This pinpoint test is intended to diagnose the following:

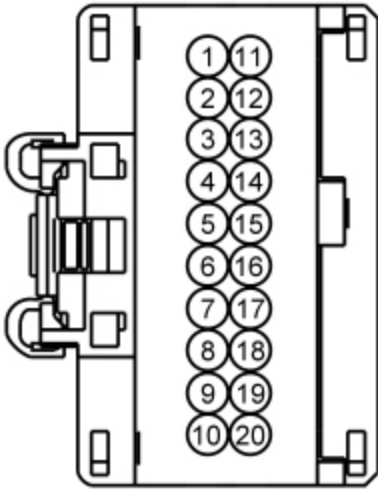
- Wiring, terminals or connectors
- HVAC module

PINPOINT TEST K : THE AMBIENT LIGHTING IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.	
K1 CHECK FOR COMMUNICATION TO THE HVAC MODULE	
• Ignition ON. • Enter the following diagnostic mode on the scan tool: Network Test . • Carry out the network test using the scan tool.	
Does the scan tool communicate with the HVAC module?	
Yes	GO to K2 .
No	REFER to Section 418-00.
K2 CHECK THE CLIMATE CONTROL OPERATION	
• Check the operation of the climate control system and verify that all of the climate controls operate correctly.	
Does the climate control system operate correctly?	
Yes	GO to K3 .
No	REFER to Section 412-00.
K3 CHECK THE AMBIENT LIGHTING GROUND CIRCUIT FOR AN OPEN	
• Ignition OFF.	

- Disconnect: Negative Battery Cable .
- Disconnect: HVAC Module [C228C](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C228C Pin 8	GD133 (BK)	—	Ground



•

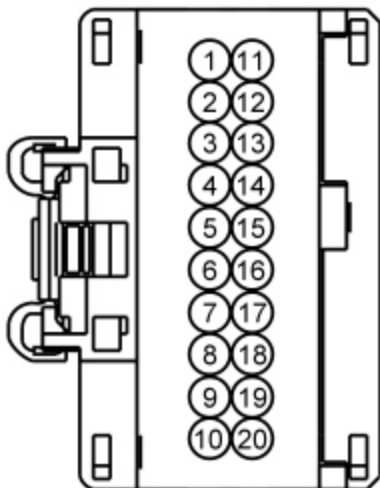
Is the resistance less than 3 ohms?

Yes	GO to K4 .
No	REPAIR circuit GD133 (BK) for an open. TEST the system for normal operation.

K4 CHECK THE AMBIENT LIGHTING RETURN CIRCUIT FOR AN OPEN

- Measure the **resistance** between:

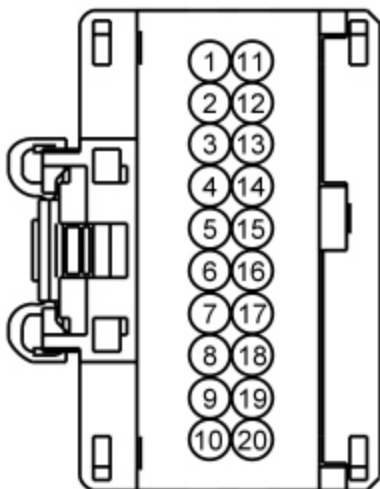
Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C228C Pin 5	VLN46 (BU)	C228C Pin 17	RLN44 (BN)



-
- **NOTE:** This step will reverse the test leads, checking the diode.

Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
C228C Pin 17	RLN44 (BN)	C228C Pin 5	VLN46 (BU)



-

Does the resistance measure less than 3 ohms one way and geater than 10,000 ohms the other way?

Yes	GO to K5 .
No	REPAIR circuit RLN44 (BN) for an open. TEST the system for normal operation.

K5 CHECK FOR CORRECT HVAC MODULE OPERATION

- Disconnect all the HVAC module connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins

- Connect the HVAC module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new HVAC module. REFER to Section 412-00. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test L: The Ambient Lighting Does Not Cycle Through All Color Combinations

Refer to Wiring Diagrams Cell [89](#) , Interior Lamps for schematic and connector information.

Normal Operation

The HVAC module provides voltage and ground to the LEDs. There are 3 LEDs (red, blue and green) housed within each LED assembly. By illuminating various color combinations, the LEDs are able to produce 7 different colors of ambient light.

The HVAC module receives input from the Accessory Protocol Interface Module (APIM) and the Body Control Module (BCM) over the communication network for color and brightness settings. The HVAC module retains the last color and brightness setting between uses.

There ambient lighting system illuminates the following:

- Cupholders (located in the center console)
- Center console storage compartment
- Front footwells
- Rear footwells

Based on input from the APIM and the BCM , the HVAC module provides voltage to the red, green and/or blue output circuits to produce the color selected.

DTC Description	Fault Trigger Conditions
• B127C:11 — Ambient Lighting Zone 2 Output Red LED: Circuit Short To Ground	A continuous and on-demand DTC that sets when the HVAC module detects a short to ground from the zone 2, red ambient lighting output circuit.
• B127C:12 — Ambient Lighting Zone 2 Output Red LED: Circuit Short To Battery	A continuous and on-demand DTC that sets when the HVAC module detects an open from the zone 2, red ambient lighting output circuit.
• B127C:13 — Ambient Lighting Zone 2 Output Red LED: Circuit Open	A continuous and on-demand DTC that sets when the HVAC module detects an open from the zone 2, red ambient lighting output circuit.
• B127F:11 — Ambient Lighting Zone 2 Output Green LED: Circuit Short To Ground	A continuous and on-demand DTC that sets when the HVAC module detects a short to ground from the zone 2, green ambient lighting output circuit.

DTC Description	Fault Trigger Conditions
B127F:11 — Ambient Lighting Zone 2 Output Green LED: Circuit Short To Battery	A continuous and on-demand DTC that sets when the HVAC module detects an open from the zone 2, green ambient lighting output circuit.
B127F:13 — Ambient Lighting Zone 2 Output Green LED: Circuit Open	A continuous and on-demand DTC that sets when the HVAC module detects an open from the zone 2, green ambient lighting output circuit.
B1282:11 — Ambient Lighting Zone 2 Output Blue LED: Circuit Short To Ground	A continuous and on-demand DTC that sets when the HVAC module detects a short to ground from the zone 2, blue ambient lighting output circuit.
B1282:12 — Ambient Lighting Zone 2 Output Blue LED: Circuit Short To Battery	A continuous and on-demand DTC that sets when the HVAC module detects an open from the zone 2, blue ambient lighting output circuit.
B1282:12 — Ambient Lighting Zone 2 Output Blue LED: Circuit Open	A continuous and on-demand DTC that sets when the HVAC module detects an open from the zone 2, blue ambient lighting output circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- HVAC module

PINPOINT TEST L : THE AMBIENT LIGHTING DOES NOT CYCLE THROUGH ALL COLOR COMBINATIONS

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

L1 CHECK THE DISPLAY INTERFACE OPERATION

- Ignition ON.
- Check the operation of the HVAC system using the buttons on the touch screen.

Does the HVAC system operate correctly?

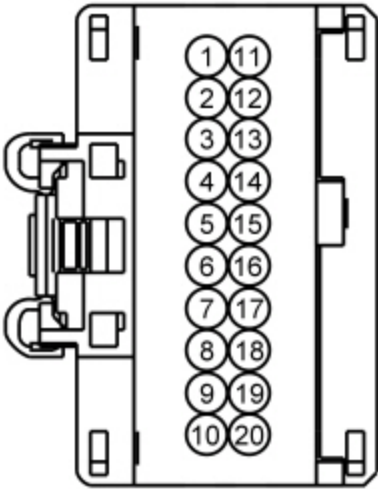
Yes	GO to L2 .
No	REFER to Section 412-00.

L2 CHECK THE AMBIENT LIGHTING HARNESS USING DIODE CHECK

- Ignition OFF.
- Disconnect: HVAC Module [C228C](#).
- Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit

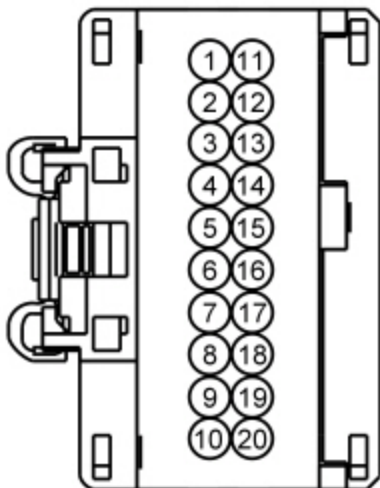
Red LED Color			
C228C Pin 7	VLN44 (OG)	C228C Pin 17	RLN44 (BN)
Green LED Color			
C228C Pin 6	VLN45 (GN)	C228C Pin 17	RLN44 (BN)
Blue LED Color			
C228C Pin 5	VLN46 (BU)	C228C Pin 17	RLN44 (BN)



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- **NOTE:** *This step will reverse the test leads, checking the diode.*

Measure the **resistance** between:

Positive Lead		Negative Lead	
Pin	Circuit	Pin	Circuit
Red LED Color			
C228C Pin 17	RLN44 (BN)	C228C Pin 7	VLN44 (OG)
Green LED Color			
C228C Pin 17	RLN44 (BN)	C228C Pin 6	VLN45 (GN)
Blue LED Color			
C228C Pin 17	RLN44 (BN)	C228C Pin 5	VLN46 (BU)



Does the resistance measure less than 3 ohms one way and geater than 10,000 ohms the other way?

Yes	GO to L3 .
No	INSTALL a new ambient lighting harness. TEST the system for normal operation.

L3 CHECK FOR CORRECT HVAC MODULE OPERATION

- Disconnect all the HVAC module connectors.
- Check for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect the HVAC module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new HVAC module. REFER to Section 412-00. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.